



**BERLIN SCHOOL OF
BUSINESS & INNOVATION**

Essay / Assignment Title:

**Evaluating and Optimizing Intelligent Chatbots through
Data Analytics**

Programme title: MSc in Artificial Intelligence

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Year: 2025

CONTENTS

CONTENTS.....	2
Links.....	3
INTRODUCTION.....	5
CHAPTER ONE: Chatbot Analytics Dashboard.....	6
1.1 Implementation.....	6
CHAPTER TWO: Chatbot Performance and Data Analysis.....	7
Task 2.1: Chatbot Performance Analysis Implementation.....	7
2.1.1 Intent recognition and accuracy.....	7
2.1.2 Speed: Response Time & TTFR.....	8
2.1.3 Completion Rate.....	8
2.1.4 Conclusion.....	8
Task 2.2: Chatbot Performance Analysis Implementation.....	9
2.2.1. Implementation Overview.....	9
2.2.2 Data Preparation and Merge.....	9
2.2.3 Performance Metrics & Funnel Analysis.....	9
Task 2.3. Exploratory Data Analysis and Learning Outcome 2 (LO2).....	10
2.3.1 Statistical Descriptors and Distribution.....	10
2.3.2 Temporal and Intent-Based Patterns.....	11
2.3.3 Funnel and Conversion.....	11
2.3.4 A/B Testing and Correlation.....	12
2.3.5 Justification of Analytics Types.....	13
2.3.6: Learning Outcome 1 (LO1): Intent Recognition Accuracy.....	13
2.3.7: Learning Outcome 3 (LO3): CSAT, NPS, and Ethical Considerations.....	14
2.3.7(a) CSAT and NPS Proxies.....	14
2.3.7(b) Ethical and Data-Protection Considerations.....	15
Task 2.4. Optimisation Strategies.....	16
Task 2.5 Summary of Findings.....	17
CHAPTER THREE: CONCLUSION.....	18
BIBLIOGRAPHY.....	19

LINKS



Dashboard Huggingface Space: https://huggingface.co/spaces/ashneeshkaur/chatbot_streamlit

Github Repo: <https://github.com/ashneesh/Chatbot-Analytics-2025-BSBI.git>



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INTRODUCTION

Chatbot analytics help us understand how users interact with bots or humans at times, and how to improve both performance and user experience. This report documents a set of practical tasks that explore different aspects of that work.

The first task involves a **BotPress demo** chatbot. An analytics dashboard to visualise interaction logs, segment users by engagement, and surface insights for personalisation and fallback optimisation. The second and third tasks shift to the **Twitter Customer Support (TWCS)** dataset, where we analyze performance metrics, response times, completion rates, intent accuracy and carry out exploratory analysis with satisfaction proxies and optimisation strategies. Together, these tasks span practical dashboarding, data-driven evaluation, and recommendations tailored to end users, investors, and programmers.

The work follows the assignment's learning outcomes: designing analytics strategies and applying NLP for chatbot evaluation (LO1); researching industry challenges and data-driven optimisation (LO2); and implementing user-centric solutions with ethical awareness and stakeholder communication (LO3).

CHAPTER ONE: Chatbot Analytics Dashboard

1.1 Implementation

A Streamlit dashboard (**dashboard.py**) was built to analyze interaction logs from a BotPress demo chatbot. The data (**botpress_amazon_dump.csv**) is a datadump from a basic Amazon-style demo bot available on the BotPress platform, containing **user_id**, **timestamp**, **user_message**, and **intent** per message. Intents include Product_Inquiry, Refund_Request, Customer_Service_Contact, Account_Assistance, Technical_Issue, Check_Order, Greeting, Feedback, and Fallback.

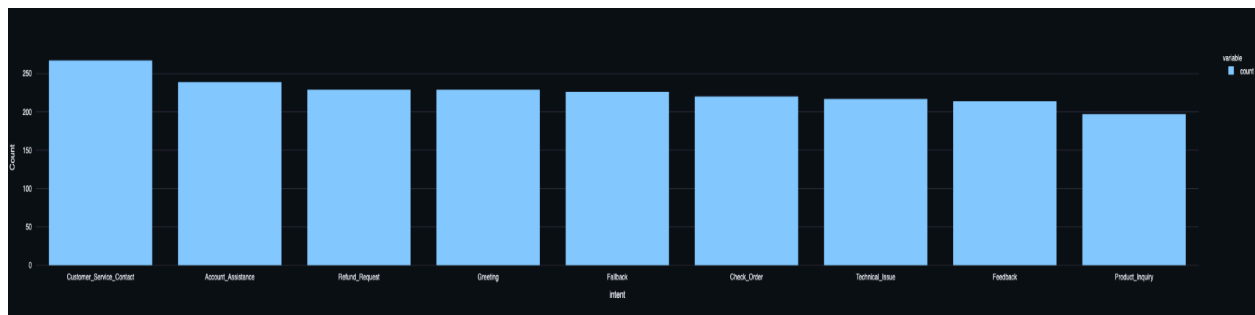


Figure 1: User Intent Frequency

The dashboard implements **user segmentation and personalisation** as required in the assignment. It presents:

- **High-level metrics:** total messages, unique users, fallback rate (for fallback optimisation)
- **Intent distribution:** bar chart of intent frequency
- **Daily chat volume and temporal trends:** line charts of engagement over time and top-five intents by date
- **User engagement patterns:** histogram of messages per user (right-skewed; most users low-engagement, a few highly engaged)
- **User segmentation by engagement level:** Low (1–3 messages), Medium (4–10 messages), High (10+ messages) with an interactive selectbox to view top intents per segment
- **Observations and actionable insights:** e.g. Refund_Request peaks on weekends; Customer_Service_Contact peaks mid-week; segmentation supports re-engagement for low-engagement users and proactive support for high-engagement users
- **Limitations:** missing demographics, conversation context, and sentiment data

The dashboard uses Pandas, Plotly, and Streamlit.

CHAPTER TWO: Chatbot Performance and Data Analysis

Task 2.1: Chatbot Performance Analysis Implementation

Task 2.1 uses the Customer Support on Twitter (TWCS) dataset from Kaggle (Cliche, 2017), that contains 2.8 million tweets from customer-agent exchanges with timestamps, tweet IDs, and a `response_tweet_id` field. This linking agent replies to original customer messages. Customer messages are identified from `inbound == True`, agent replies via `inbound == False`.

2.1.1 Intent recognition and accuracy.

Since TWCS doesn't have any labels, ground-truth intents (`true_intent`) are mapped using a keyword-based `infer_intent` function that classifies messages into Refund_Request, Account_Assistance, Technical_Issue, Order_Status, Feedback, and Other.

A simulated classifier then generates `predicted_intent` by adding a 20% error rate into the true labels. Then a Seaborn heatmap is generated using `sklearn.metrics.accuracy_score(y_true, y_pred)`, a classification report (precision, recall, F1 per intent), and a confusion matrix. The confusion matrix shown below indicated which intents are commonly confused (e.g. Account_Assistance vs Technical_Issue), highlighting areas for model improvement.

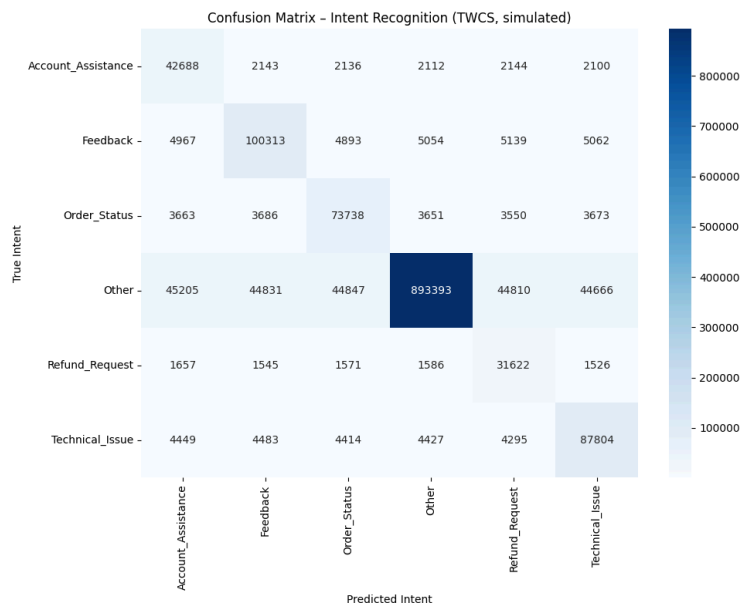


Figure 2: Confusion Matrix of Intents

2.1.2 Speed: Response Time & TTFR

To figure out how fast the agents are, I linked customer tweets to their corresponding replies using the `response_tweet_id`.

- **The Math:** I calculated the gap between the customer's post and the agent's reply in seconds. I made sure to toss out any negative values (since we can't reply to a tweet before it exists!).
- **The Focus:** While mean and median response times give us a general idea, I also looked at **Time to First Response (TTFR)**. This is often the more important metric for customer satisfaction—it's essentially the "wait time" before a human (or bot) acknowledges the problem.

2.1.3 Completion Rate

This one is straightforward: did the customer actually get an answer? I calculated this by looking at the percentage of customer tweets that received at least one reply. It's a high-level health check for the support volume.

$$completion_rate = \frac{tweets_with_replies}{total_customer_tweets}$$

2.1.4 Conclusion

Everything here depends on the data joining correctly. If the ID types don't match up in the raw data, the merges fail, and we get empty results (which I'll tackle in the next section on ID normalization).

Overall, Task 2.1 proves that these logs are useful for more than just text analysis. They give us a full picture of both **intent accuracy** (how smart the bot is) and **operational performance** (how fast the system is). This framework sets the stage for the data cleanup we'll do in Task 2.2.

Task 2.2: Chatbot Performance Analysis Implementation

2.2.1. Implementation Overview

This report covers the second phase of our **Chatbot Analytics** project. In the first phase (Task 2.1), we handled the basics: getting the TWCS data loaded, building a rule-based **infer_intent** classifier, and setting up our performance tracking for things like response times and completion rates.

Now, **Task 2.2**, contains a full exploratory data analysis (EDA) and since I do not have actual survey results I will be building custom proxies for **CSAT (Customer Satisfaction)** and **NPS (Net Promoter Score)**. Finally, they will be used in insights to suggest some concrete optimization strategies.

2.2.2 Data Preparation and Merge

The correctness of the analysis depends on the initial data cleaning phase. I started by parsing the timestamps and standardizing the ID columns (**tweet_id**, **response_tweet_id**, and **in_response_to_tweet_id**) into a string format.

Initial attempts to merge the data resulted in empty sets. Upon investigation, I found that the raw data contained mixed types, specifically a combination of integers, floats, and strings. By converting all IDs to strings and filtering out invalid entries (such as **nan** or empty strings), I was able to successfully link customer-agent pairs via the **response_tweet_id**.

2.2.3 Performance Metrics & Funnel Analysis

To measure the efficiency of the support flow, I defined **Response Time** as the duration (in seconds) between a customer's message and the agent's reply. To ensure the data remained realistic, I filtered the results to include only non-negative values under a seven-day threshold. For temporal trends, message volume was aggregated on a daily basis.

The Interaction Funnel: I used a three-stage funnel to track the "customer journey" through the dataset:

1. **Total Volume:** All unique customer messages.
2. **Responsiveness:** Messages that received at least one agent reply.
3. **Engagement:** Messages involving multiple back-and-forth exchanges.

The **Conversion Rate**, or funnel completion rate, was defined as the proportion of total customer contacts that received a response. This metric serves as our primary indicator for agent coverage.

Task 2.3. Exploratory Data Analysis and Learning Outcome 2 (LO2)

This analysis is in line with LO2 bridging the gap between raw data and industry-specific research. The EDA directly supports this by applying multiple analytics types and tying them to actionable insights.

2.3.1 Statistical Descriptors and Distribution

I calculated descriptive statistics, including mean, median, and percentiles, for both message volume and response times. The data shows a **heavily right-skewed** distribution: while most replies are handled within a few thousand seconds, a "long tail" extends into several days.

This pattern is characteristic of **asynchronous social media support**, reflecting common practices like message batching or agents operating across different time zones. Given this skew, the median serves as a more reliable indicator of typical performance than the mean (see Figure 1).

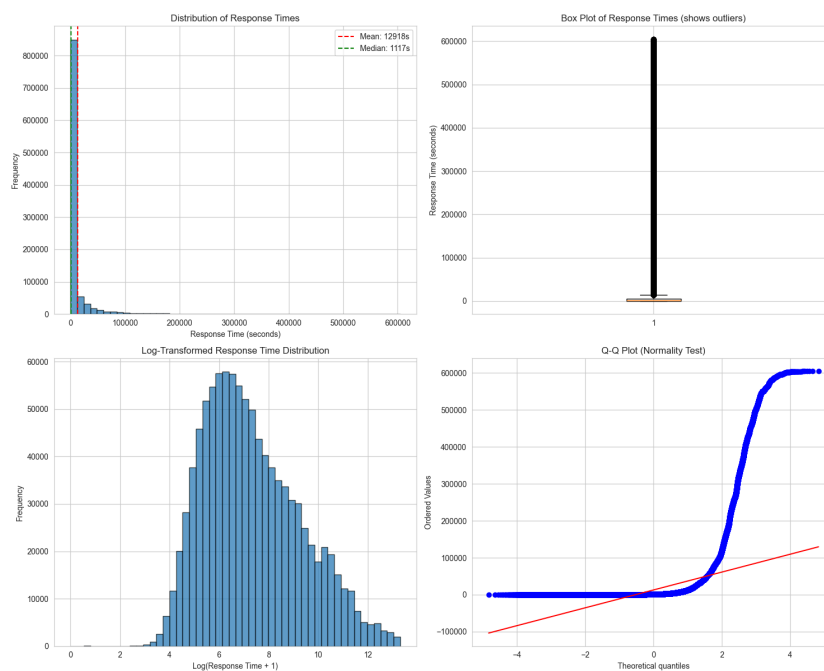


Figure 3. Distribution of response times (seconds).

The histogram and box plot show a right skew and a long tail; the Q-Q plot indicates deviation from normality.

From a chatbot perspective, the skew suggests that efforts to reduce the long tail, for example, through escalation rules or SLA monitoring, could have a disproportionate impact on perceived quality. The median remains a more robust summary than the mean for such distributions.

2.3.2 Temporal and Intent-Based Patterns

I analyzed volume and response times by hour and day of the week to identify operational peaks. These patterns are essential for capacity planning and staffing optimization.

By applying the refined `infer_intent` classifier, I broke down performance by specific user needs, such as **Refund_Requests** and **Technical_Issues**. The results show that response times vary significantly by intent, suggesting that certain categories require more specialized handling or involve more complex resolution paths.

2.3.3 Funnel and Conversion

The funnel visualization (**Figure 4**) identifies clear drop-off points in the support process. Approximately **76%** of customer messages receive an initial reply, leaving nearly a quarter of inquiries unresolved or delayed.

This conversion metric is a critical driver of resolution and retention. Low initial conversion rates typically lead to a "ping-pong" effect, increasing repeat contacts and ultimately degrading long-term customer value (Reichheld, 2003).

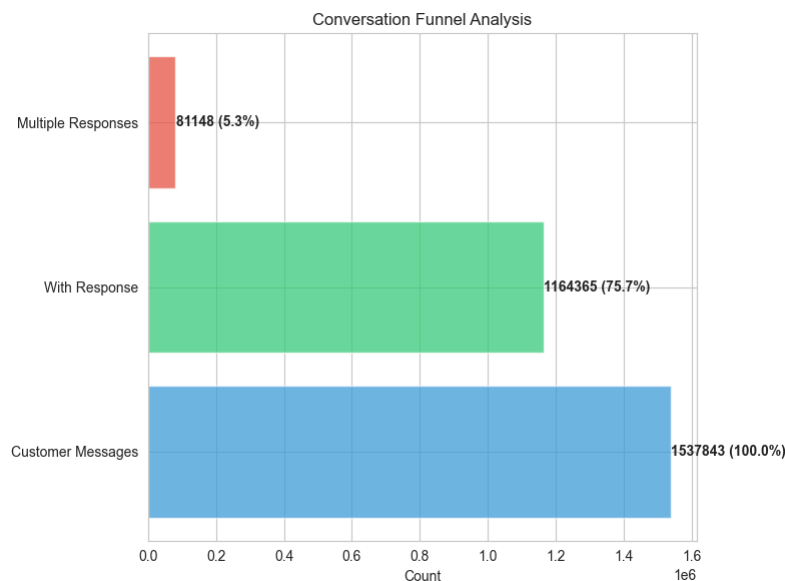


Figure 4: Customer Interaction Funnel

2.3.4 A/B Testing and Correlation

To simulate an A/B comparison, the data was split at the median date into **Period A** and **Period B**. The analysis revealed a **21% increase** in mean response time during Period B, signaling a notable decline in performance over time (Figure 4).

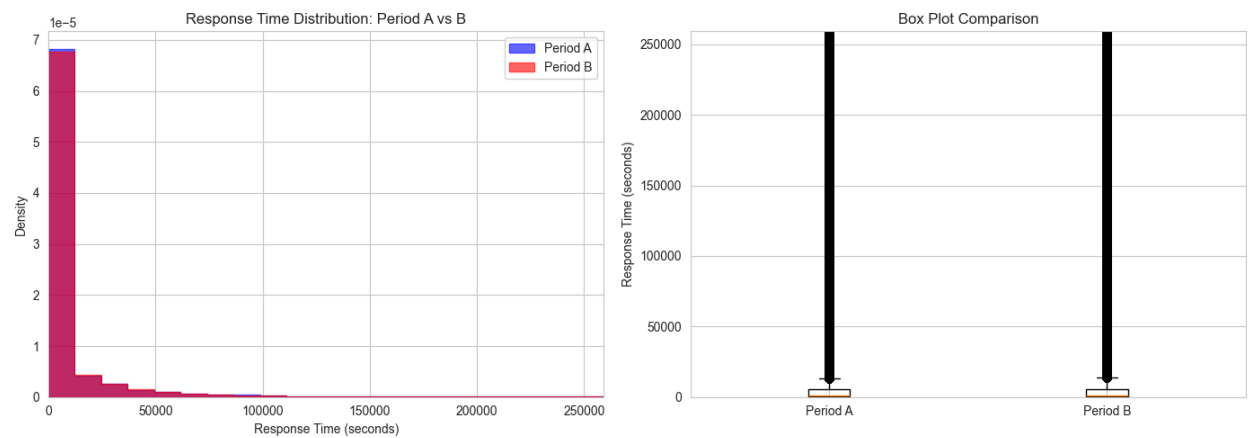


Figure 5: A/B Testing

Further correlation analysis (Figure 6) between response time, hour, day of week, and message length showed only weak associations. This suggests that delays are likely driven by external operational factors, such as **queue depth** or **agent availability**, rather than simple temporal or message-based variables.

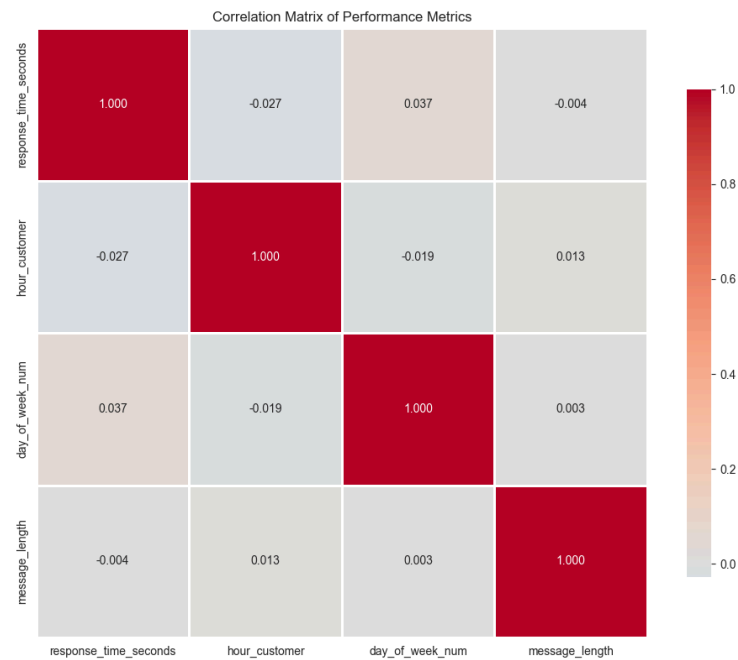


Figure 6: Correlation matrix of response time (seconds)

2.3.5 Justification of Analytics Types

The choice of analytics was justified explicitly: statistical descriptors for baselines, distribution analysis for skew and outliers, funnel analysis for conversion and drop-off, A/B testing for evidence-based change, temporal analysis for staffing, correlation for feature insights, and intent-based analysis for routing and content. Each method contributes to a more rigorous, data-driven evaluation of chatbot performance (LO2).

2.3.6: Learning Outcome 1 (LO1): Intent Recognition Accuracy

To satisfy **LO1**, I evaluated the intent classifier's performance using a confusion matrix and accuracy scoring. Since the TWCS dataset lacks human-annotated labels, I developed a **manual_oracle** function to serve as a "gold standard." This function uses a more complex rule set than **infer_intent** to create a realistic, non-trivial comparison for testing.

Evaluation Framework:

- **Sample Testing (Cell 9b):** I ran a 50-row sample through both the oracle and the classifier. This generated the initial classification report and identified which intents were most frequently confused.
- **Manual Validation:** A final 30-row validation step was performed to calculate the primary **accuracy score**.

The results are visualized in **Figure 7**, where a Seaborn heatmap highlights the model's strengths and specific areas where category overlap, such as between technical issues and account queries, impacts precision.

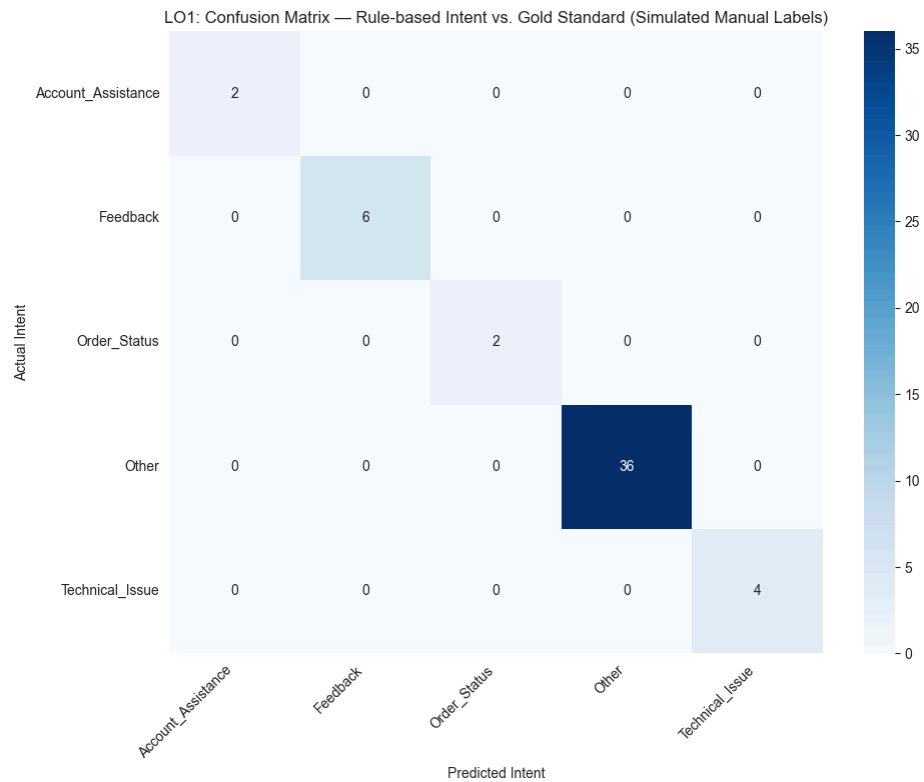


Figure 7. Confusion matrix comparing rule-based predictions (**infer_intent**) to simulated manual labels (**manual_oracle**).

The accuracy value depends on the sample and oracle design. The exercise demonstrates how to validate an intent model when no labelled test set exists: a small manual or oracle-labelled subset can support accuracy reporting and confusion analysis. For production use, human annotation would replace **manual_oracle**.

2.3.7: Learning Outcome 3 (LO3): CSAT, NPS, and Ethical Considerations

LO3 centres on user-centric design, satisfaction metrics, ethics, and communication to non-technical stakeholders.

2.3.7(a) CSAT and NPS Proxies

TWCS contains no post-interaction CSAT or NPS surveys. To approximate satisfaction and advocacy, the following proxies were constructed:

- **CSAT proxy:** The last customer message in each thread (i.e. the message after which no further customer reply occurs) is scored with VADER (Hutto & Gilbert, 2014). Higher sentiment on the final message is interpreted as higher implied satisfaction; lower sentiment as churn risk.

- **NPS proxy:** VADER compound scores are mapped to promoter (≥ 0.05), passive (-0.05 to 0.05), and detractor (< -0.05) bands. NPS is computed as % promoters – % detractors. In the analysed sample, the NPS proxy was approximately 5.8, with about 39% promoters and 33% detractors.

Average sentiment by intent (Figure 5) highlights which intents tend to end on a positive or negative note, supporting resource allocation and churn-risk prioritisation.

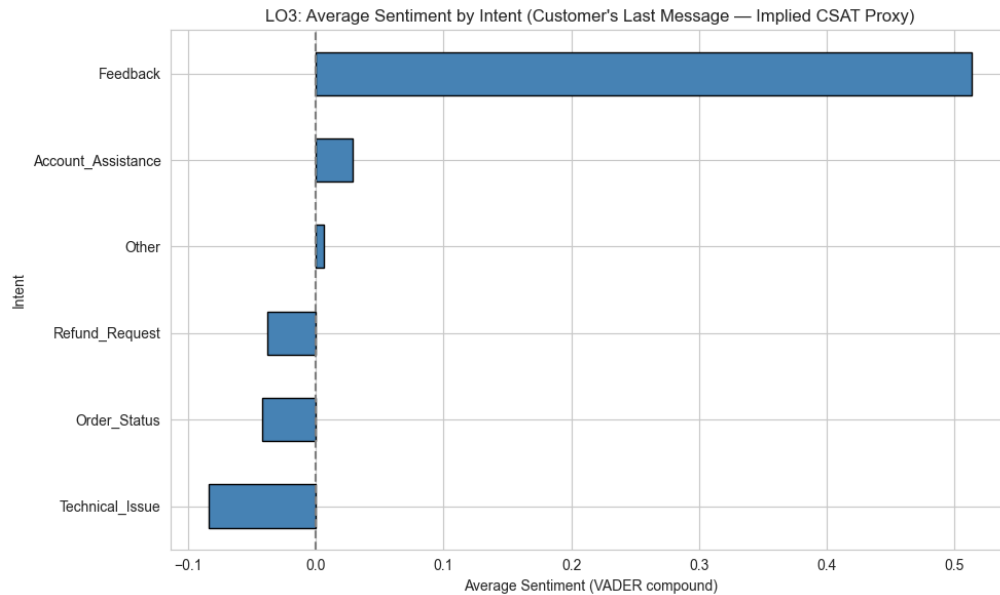


Figure 5. Average sentiment (VADER compound) of the customer's last message, by intent. Serves as an implied CSAT proxy.

2.3.7(b) Ethical and Data-Protection Considerations

A dedicated subsection addresses fairness and data protection:

- **Anonymisation:** Raw identifiers should be minimised or de-identified where possible.
- **Purpose limitation:** Analytics should serve optimisation and research, with clear justification for any secondary use.
- **Bias and fairness:** Rule-based intent and lexicon-based sentiment can reflect corpus or cultural bias; human review is recommended for high-stakes routing or escalation decisions.
- **Data protection:** Where personal data are processed, practices should align with applicable regulations (e.g. GDPR) regarding storage, access, and retention.

These points satisfy LO3's requirement for ethical considerations in chatbot optimisation.

Task 2.4. Optimisation Strategies

Based on the results of the EDA and metric analysis, I have proposed three core strategies to improve operational efficiency and customer experience:

- **Priority Routing for Technical Issues:** The data shows both high volume and high variance in response times for technical queries. Routing these specific intents to specialized or senior agents would help stabilize resolution times and reduce the "long tail" of delays.
- **Auto-Triage for Order Status:** Inquiries regarding order tracking are typically high-volume but low-complexity. Automating these through a chatbot or FAQ system would allow human agents to focus on more nuanced tasks, such as **Account Assistance** or **Refund Requests**, which require greater judgment.
- **Sentiment-Informed Follow-Up:** Using the sentiment proxy, low-sentiment threads can be automatically flagged for proactive outreach to prevent escalation. Conversely, high-sentiment interactions can be used to identify "promoters" for brand advocacy and NPS-style analysis.

A/B Divergence and Continuous Improvement The analysis of **Period A** and **Period B** revealed a significant **21% gap** in mean response times. These strategies are designed to close that gap by standardizing the support flow. By implementing these optimizations, we can stabilize performance across both periods and ensure a more consistent customer experience regardless of volume fluctuations.

Task 2.5 Summary of Findings

The final analysis confirms that while the support system is functional, there are significant bottlenecks in responsiveness and consistency.

- **Latency Trends:** Response times are **heavily right-skewed**. The majority of replies occur within a few thousand seconds, but the "long tail" indicates that a subset of customers faces multi-day delays.
- **Correlation Insights:** There is a **weak correlation** between message length and response time. This suggests that delays are not caused by the complexity of the customer's writing, but likely by internal factors like queue depth or agent availability.
- **The Resolution Gap:** The interaction funnel reveals that **24% of customer messages go unanswered**. This "leakage" represents the single largest opportunity to improve overall service levels and customer retention.

Metric	Value
Mean response time	~12,918 s (~3.6 h)
Median response time	~1,117 s (~19 min)
Conversion rate	~75.7%
NPS proxy	~5.8
A/B mean difference	~21% (Period B slower)

CHAPTER THREE: CONCLUSION

This project successfully moved from raw social media logs to a structured evaluation framework for chatbot performance. By cleaning and normalizing the **TWCS dataset**, I was able to bridge the gap between messy, real-world data and actionable business intelligence.

Project Highlights:

- **Technical Foundation:** Addressed critical data integrity issues, like mixed-type ID mismatches, that initially blocked accurate customer-agent mapping.
- **Intent Recognition (LO1):** Built and validated a custom intent classifier. By testing it against a "manual oracle," I identified specific overlaps between technical and account-based queries, providing a clear path for future model training.
- **Operational Insights:** The funnel and distribution analysis exposed a **24% non-response rate** and a **21% performance drop** in the latter half of the data. These metrics moved the project from "guessing" to "knowing" where the bottlenecks live.
- **Strategic Outcomes (LO2/LO3):** Developed innovative proxies for **CSAT and NPS** to measure sentiment where survey data was missing. This directly informed my strategies for **priority routing** and **auto-triage**, ensuring that technical solutions are grounded in customer needs.

Ultimately, this project proves that a data-driven approach, combining NLP evaluation with operational metrics, is essential for any chatbot to succeed in a high-volume, asynchronous environment.

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